

Positions

- 2024-now **Postdoctoral researcher**, *Inria Universit   de C  te d'Azur - PreMeDICAL team*, Montpellier, France
- 2021-2024 **Ph.D. student**, *Inria Saclay - Mind team*, Palaiseau, France
- April 2021 - **Internship**, *Inria Saclay*, Palaiseau, France
- September 2021 ○ Advisors: Samuel Vaiter, Thomas Moreau and Pierre Ablin
- 2021 ○ Subject: Stochastic bilevel optimization for hyperparameter selection
- May 2020 - **Internship**, *EDF R&D*, Chatou, France
- November 2020 ○ Advisors: Alexandre Girard, Yannig Goude, Giorgio Simonini
- 2020 ○ Subject: Machine learning for nuclear unit control

Education & Diplomas

- 2025 **CNU Qualification for Associate Professor (MCF)**, *Section 26 (Applied Mathematics)*
- 2021-2024 **Ph.D. student in Mathematics & Computer Science**, *Inria Saclay & Universit   Paris-Saclay*, Palaiseau, France
- 2021-2024 ○ Advisors: Samuel Vaiter, Thomas Moreau and Pierre Ablin
- 2021-2024 ○ Subject: Contributions to stochastic bilevel optimization
- 2020-2021 **M.Sc. Mathematics, Vision, Learning**, *  cole Normale Sup  rieure Paris-Saclay*, Gif-Sur-Yvette, France
- 2019-2020 **First year in Master's degree in Mathematics**, *Sorbonne Universit  *, Paris, France
- 2019-2020 Remote track - Probabilities, statistics, dynamic systems, functional analysis, stochastic calculus and stochastic control
- 2018-2019 **Bachelor's degree in Mathematics**, *Sorbonne Universit  *, Paris, France
- 2018-2019 Remote track, with highest honour
- 2017-2020 **Engineering degree**, *  cole Centrale de Nantes*, Nantes, France
- 2014-2017 **Classes pr  paratoires**, *Lyc  e Michel Montaigne*, Bordeaux, France

Publications

International Conferences

1. L. Bleistein, **M. Dagr  ou**, F. de Lima Andrade*, T. Boudou, and A. Bellet. *Optimal Transport under Group Fairness Constraints*. In *Advances in International Conference on Machine Learning (ICML)*, **Spotlight (Top 2.2%)**, 2026.
2. **M. Dagr  ou**, T. Moreau, S. Vaiter., P. Ablin. A Lower Bound and a Near-Optimal Algorithm for Bilevel Empirical Risk Minimization. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
3. **M. Dagr  ou**, P. Ablin, S. Vaiter., T. Moreau. A framework for bilevel optimization that enables stochastic and global variance reduction algorithms. In *Advances in Neural Information Processing Systems (NeurIPS)*, **Oral equivalent paper (Top 2%)**, 2022.
4. T. Moreau, M. Massias, A. Gramfort, Pierre Ablin, P.-A. Bannier, B. Charlier, **M. Dagr  ou**, T. Dupre la Tour, G. Durif, C. F Dantas, Q. Klopfenstein, J. Larsson, E. Lai, T. Lefort, B. Mal  zieux, B. Moufad, B. T Nguyen, A. Rakotomamonjy, Z. Ramzi, J. Salmon, S. Vaiter. Benchopt: Reproducible, efficient and collaborative

optimization benchmarks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

National Conferences

1. **M. Dagr  ou**, T. Moreaux, S. Vaiter, P. Ablin. Borne inf  rieure de complexit   et algorithme quasi-optimal pour la minimisation de risque empirique bi-niveaux. In *XXIX  me Colloque Francophone de Traitement du Signal et des Images GRETSI*, 2023.
2. **M. Dagr  ou**, P. Ablin, S. Vaiter, T. Moreau. Algorithmes stochastiques et r  duction de variance gr  ce    un nouveau cadre pour l'optimisation bi-niveaux. In *XXVIII  me Colloque Francophone de Traitement du Signal et des Images GRETSI*, 2022.

Workshop paper

1. L. Bleistein, **M. Dagr  ou**, F. de Lima Andrade, T. Boudou, and A. Bellet. *Optimal Transport under Group Fairness Constraints*. EurIPS workshop *Unifying Perspectives on Learning Biases*. 2025, <https://openreview.net/forum?id=FAi7sNLS9w>

Miscellaneous

1. **M. Dagr  ou**, P. Ablin, S. Vaiter, and T. Moreau. How to compute Hessian-vector products?, In *ICLR blogpost track, Highlight (top 10%)*, 2024, <https://iclr-blogposts.github.io/2024/blog/bench-hvp/>

Awards

- 2026 **Gold Reviewer**, *ICML 2026*, (Top 25%), <https://icml.cc/Conferences/2026/ProgramCommittee>
- 2023 **Top Reviewer**, *NeurIPS 2023*, (Top 10%), <https://neurips.cc/Conferences/2023/ProgramCommittee>
- 2023 **TICS Doctoral School of Paris-Saclay prize**, <https://www.universite-paris-saclay.fr/ecoles-doctorales/sciences-et-technologies-de-linformation-et-de-la-communication-stic>

Teaching

- 2023 **Optimization**, *CentraleSupélec*, Teaching assistant
 -    2nd year of engineering program at CentraleSup  lec, Master's level (M1)
 -    Supervision of tutorial sessions and practical labs on Matlab, grading of assignments

Reviewing service

Journals

-    Transactions on Machine Learning Research (2025)
-    Journal of Machine Learning research (2023, 2024, 2025)
-    IEEE Signal Processing Magazine (2023)
-    Machine Learning (2022)

Conferences

-    International Conference on Learning Representations (2026)
-    Neural Information Processing Systems (2023, 2024, 2025, 2026)
-    International Conference on Machine Learning (2023, 2024, 2025, 2026)
-    Conference on Artificial Intelligence and Statistics (2024, 2026)

Workshop

-    Privacy-Preserving Machine Learning Workshop at EurIPS 2025 (2025)

Communication

- 2026-03 Talk at Institut de Math  matiques de Toulouse (Toulouse): *Bilevel optimization for machine learning*

- 2026-01 Talk at Institut de Mathématiques de Bordeaux (Bordeaux): *Bilevel optimization for machine learning*
- 2026-01 Talk at MAGNET team seminar (Lille): *Bilevel optimization for machine learning*
- 2025-02 Talk at Machine Learning in Montpellier, Theory & Practice (Montpellier): *Bilevel optimization for machine learning*
- 2024-06 Talk at STIC doctoral day (Gif-sur-Yvette): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*
- 2024-05 Poster Session at AISTATS (Valencia): *A lower bound a near-optimal algorithm for bilevel empirical risk minimization*
- 2023-09 Poster Session at GRETSI (Grenoble): *A lower bound a near-optimal algorithm for bilevel empirical risk minimization*
- 2023-06 Poster Session at the workshop "Optimization and machine learning (Toulouse): *A lower bound a near-optimal algorithm for bilevel empirical risk minimization*
- 2023-02 Talk at Center of Data Science (ENS): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*
- 2022-12 Poster Session at NeurIPS (New Orleans): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*
- 2022-11 Poster Session at NeurIPS@Paris (Paris): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*
- 2022-10 Poster Session at GDR MOA (Nice): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*
- 2022-09 Poster Session at GRETSI (Nancy): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*
- 2022-06 Poster Session at Curves and Surfaces (Arcachon): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*
- 2022-04 Talk at the Parietal Meeting: *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*
- 2022-03 Talk at Proba-Stat seminar (LJAD Nice): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*
- 2022-03 Talk at the Miles team seminar (LAMSADE): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*

Computer skills

Language Python
Tools Unix, Git, $\text{\LaTeX}$, SLURM, OAR

Languages

French Native speaker
English Fluent
Spanish Fluent